

ROCKGLEN, SK, Canada

WMO: 711350

Lat: 49.1678N

Lon: 105.9792W

Elev: 3009

StdP: 13.17

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -20.8	(c) -15.2	(d) -28.4	(e) 1.2	(f) -20.6	(g) -22.1	(h) 1.8	(i) -15.0	(j) 31.3	(k) 23.7	(l) 28.6	(m) 22.3	(n) 9.2	(o) 320	(p) 0.712

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	88.0	62.7	84.3	61.7	80.8	60.6	66.6	79.8	64.5	77.7	62.7	76.2	11.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.4	94.3	71.5	60.1	86.7	69.1	58.0	80.3	67.7	33.0	80.5	31.2	77.3	29.8	76.2	75.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.4	23.4	21.0	-27.0	94.1	5.3	3.6	-30.8	96.7	-34.0	98.8	-37.0	100.8	-40.8	103.5
			-26.9	70.7	5.5	2.6	-30.8	72.6	-34.1	74.2	-37.1	75.7	-41.1	77.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.6	12.9	15.1	25.7	39.5	50.5	58.9	66.3	65.1	55.6	41.3	27.1	15.8		
	DBStd	21.96	15.19	13.96	13.70	10.24	8.38	6.52	6.09	7.01	9.14	9.96	12.65	13.83		
	HDD50	5414	1152	978	753	333	96	8	0	1	44	300	689	1060		
	HDD65	9502	1617	1398	1218	765	453	201	58	88	305	737	1137	1525		
	CDD50	1622	0	0	1	19	112	275	505	467	212	29	2	0		
	CDD65	232	0	0	0	0	3	18	98	90	22	1	0	0		
	CDH74	2840	0	0	0	8	76	238	1116	1074	318	10	0	0		
	CDH80	924	0	0	0	1	9	49	388	383	93	1	0	0		
(14) Wind	WSAvg	10.6	11.7	11.0	11.1	11.0	10.6	9.4	8.9	9.0	9.7	11.1	11.4	11.8		
(15) Precipitation	PrecAvg	14.0	0.6	0.4	0.7	1.1	2.2	2.6	2.1	1.5	1.0	0.9	0.5	0.7		
	PrecMax	21.4	1.8	0.8	1.8	2.2	7.4	5.5	8.4	4.2	3.8	3.9	2.0	1.9		
	PrecMin	8.7	0.1	0.0	0.0	0.2	0.4	0.0	0.6	0.1	0.1	0.1	0.0	0.1		
	PrecStd	2.8	0.4	0.2	0.4	0.6	1.7	1.1	1.6	0.9	0.8	0.9	0.4	0.4		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.3	45.8	62.0	74.0	80.8	86.1	94.1	93.6	88.3	74.9	61.9	44.9		
		MCWB	36.9	38.9	47.2	51.6	57.1	62.3	63.9	63.4	60.9	53.6	47.1	37.2		
	2%	DB	38.8	40.1	53.6	66.5	75.6	80.0	88.1	88.3	82.3	66.5	53.2	40.2		
		MCWB	34.4	35.2	42.2	49.1	56.0	61.9	63.8	61.8	57.8	50.3	42.9	34.6		
	5%	DB	36.3	36.8	47.3	61.2	70.9	76.2	84.3	84.2	77.2	61.2	47.5	36.9		
		MCWB	32.7	33.0	39.0	46.2	53.4	59.7	63.2	60.9	56.9	48.1	39.2	32.7		
	10%	DB	33.0	33.1	42.1	56.4	66.2	72.5	80.5	80.4	71.7	56.0	42.8	33.5		
		MCWB	30.3	30.3	36.3	43.8	50.9	58.0	62.0	60.1	54.5	45.9	36.7	30.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.5	39.5	47.2	53.6	60.1	68.1	70.5	67.5	63.3	55.7	48.2	37.9		
		MCDB	41.9	44.9	60.7	70.5	74.3	79.0	82.4	82.8	80.3	70.6	60.5	43.1		
	2%	WB	34.8	35.7	42.9	50.0	57.1	63.9	67.8	64.9	60.3	52.0	43.3	35.1		
		MCDB	38.3	39.4	51.7	64.9	71.4	75.4	80.9	80.2	76.5	64.0	52.2	39.6		
	5%	WB	32.8	33.1	39.4	47.2	54.8	61.5	65.6	63.2	57.7	49.1	39.7	32.8		
		MCDB	36.1	36.4	46.2	59.3	67.9	72.6	78.2	78.2	73.4	59.5	46.5	36.6		
	10%	WB	30.3	30.4	36.5	44.7	52.7	59.4	63.7	61.5	55.5	46.3	36.9	30.4		
		MCDB	33.2	33.1	41.6	55.0	63.9	70.2	76.2	75.8	70.3	55.5	42.0	33.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	15.6	15.6	16.5	20.5	21.2	19.9	22.9	23.5	21.9	18.8	15.9	14.5		
		MCDBR	17.0	16.9	23.5	28.9	28.3	26.2	28.8	29.4	30.1	27.1	22.1	16.5		
		MCWBR	13.8	12.9	14.4	15.4	14.0	12.5	11.1	11.1	12.7	14.1	13.4	12.5		
	5% WB	MCDBR	16.8	16.6	22.5	26.7	25.1	23.0	24.4	25.5	26.9	24.9	21.6	16.2		
		MCWBR	13.9	13.2	14.4	15.0	13.1	12.3	11.6	11.0	12.4	13.8	13.8	12.7		
(40) Clear-Sky Solar Irradiance	taub		0.246	0.251	0.267	0.316	0.337	0.339	0.351	0.359	0.317	0.277	0.250	0.229		
	taud		2.431	2.469	2.545	2.412	2.396	2.440	2.423	2.372	2.460	2.563	2.529	2.487		
	Ebn at Noon		267	293	306	296	292	292	286	278	282	280	266	262		
	Edh at Noon		19	24	26	34	36	35	35	35	29	22	17	16		
(44) All-Sky Solar Radiation	RadAvg		429	753	1156	1553	1857	1999	2163	1758	1286	803	464	331		
	RadStd		40	51	69	90	136	119	118	122	116	86	53	24		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page